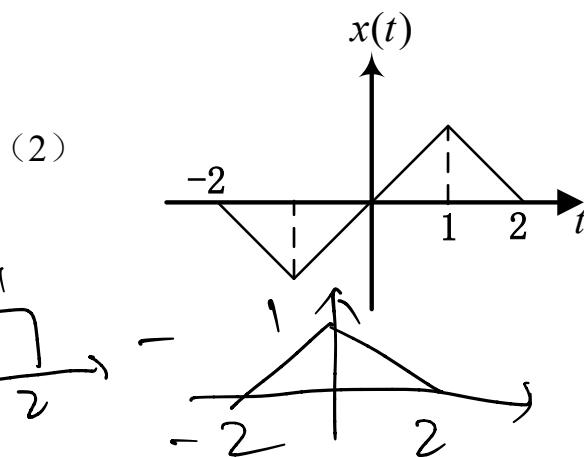
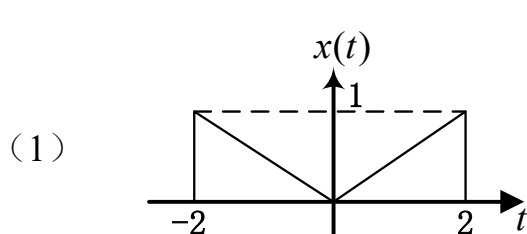


信号与系统第二次测试 (2026 年)

姓名:

学号:

1. 求如下信号的傅里叶变换

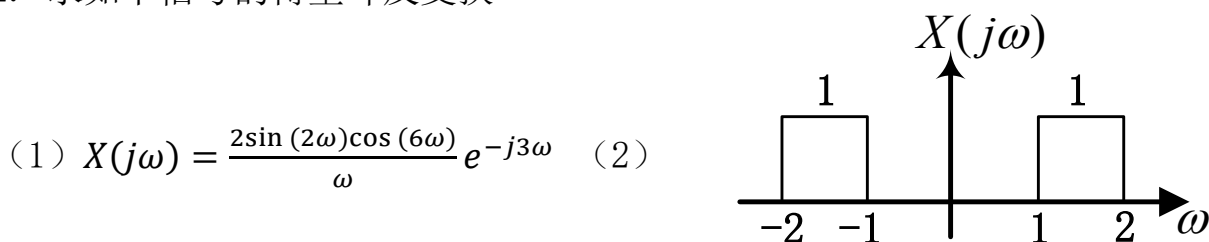


(1) 解: $x(t) =$

$$X(j\omega) = 4\text{Sa}(2\omega) - 2\text{Sa}^2(\omega)$$

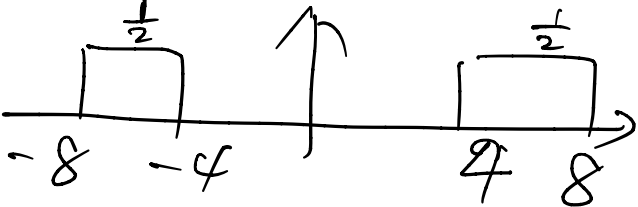
(2) 解: $X(j\omega) = \text{Sa}^2(\frac{1}{2}\omega) [e^{-j\omega} - e^{j\omega}]$
 $= -2j\text{Sa}^2(\frac{1}{2}\omega) \sin(\omega)$

2. 求如下信号的傅里叶反变换



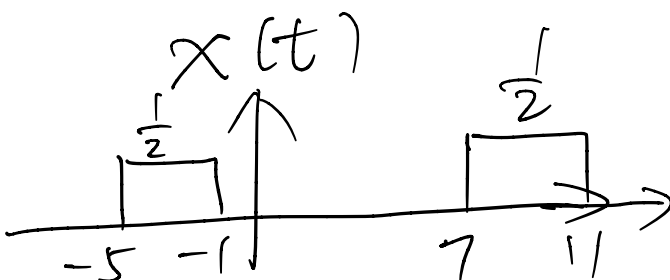
(1) 解: 设 $Y(j\omega) = \frac{2\sin(2\omega)\cos(6\omega)}{\omega}$

$$= \frac{\sin(8\omega) - \sin(4\omega)}{\omega}$$

则 $y(t) =$ 

由于 $X(j\omega) = Y(j\omega) e^{-j3\omega}$

则 $x(t) = y(t-3)$

则 

(2) $x(t) = \frac{\sin(2t) - \sin(t)}{\pi t}$

3. 已知微分方程

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

试求当 $x(t) = e^{-4t}u(t)$ 时, 系统的零状态响应 $y(t)$ 。

解: $H(j\omega) = \frac{j\omega + 1}{(j\omega + 2)(j\omega + 3)}$

$$X(j\omega) = \frac{1}{j\omega + 4}$$

$$Y(j\omega) = \frac{j\omega + 1}{(j\omega + 2)(j\omega + 3)(j\omega + 4)}$$

$$= \frac{-\frac{1}{2}}{j\omega + 2} + \frac{2}{j\omega + 3} - \frac{\frac{3}{2}}{j\omega + 4}$$

$$y(t) = \left(-\frac{1}{2}e^{-2t} + 2e^{-3t} - \frac{3}{2}e^{-4t} \right) u(t)$$